1972 Carter, building on work by Hawking and Israel, proves the no-hair conjecture for spinning, uncharged black holes (except for some technical details filled in later by Robinson). He shows that such a black hole is always described by Kerr's solution of Einstein's equation. [Ch. 7]

Thorne proposes the hoop conjecture as a criterion for when black holes form. [Ch. 7]

Bekenstein conjectures that a black hole's surface area is its entropy in disguise, and conjectures that the hole's entropy is the logarithm of the number of ways the hole could have been made. Hawking argues vigorously against this conjecture. [Ch. 12]

Bardeen, Carter, and Hawking formulate the laws of evolution of black holes in a form that is identical to the laws of thermodynamics, but maintain that the horizon's surface area cannot be the hole's entropy in disguise. [Ch. 12]

Teukolsky develops perturbation methods to describe the pulsations of spinning black holes. [Ch. 7]

1973 Press and Teukolsky prove that the pulsations of a spinning black hole are stable; they do not grow by feeding off the hole's rotational energy. [Ch. 7]

1974 Hawking shows that *all* black holes, spinning or nonspinning, radiate precisely as though they had a temperature that is proportional to their surface gravity, and they thereby evaporate. He then recants his claim that the laws of black-hole mechanics are not the laws of thermodynamics in disguise and recants his critique of Bekenstein's conjecture that a hole's surface area is its entropy in disguise. [Ch. 12]

1974–1978 Blandford, Rees, and Lynden-Bell identify several methods by which supermassive black holes in the nuclei of galaxies and quasars can create jets. [Ch. 9]

1975 Bardeen and Petterson show that the swirl of space around a spinning black hole can act as a gyroscope to maintain the